L3.3 In-class Assignment – Conditionals

CPSC 292 – 10/26/2026

Name: _____ **Section:** _____

Collaboration team: _____

Question 1

Write a simple set of `if`/`else if`/`else` statements that prints out one of three different messages that depend on the value `x`.

Question 2

You have a vector of data on spot counts for beetles that you have sampled named `spot_counts`. If the beetles have 6 spots, they are species A and if they have 10 spots, they are species B. Write a single line of code that create a new vector named `beetle_species` that determines the species automatically from the number of spots.

Question 3

You have the vectors:

```
vec1 <- c(2,1,1,3,2,1,0)
```

```
vec2 <- c(3,8,2,2,0,0,0)
```

Determine which of the following if statements would result in the string being printed to the console.

a. `if((vec1[1] + vec2[2]) == 10){ print("Print me!") }`
b. `if(vec1[1] >= 2 && vec2[1] >= 2) { print("Print me!") }`
c. `if(all((vec2 - vec1)[c(2, 6)] < 7)){ print("Print me!")}`
d. `if(!is.na(vec2[3])){ print("Print me!") }`

Question 4

Using `vec1` and `vec2` from Question 3, write a line of code that multiples the corresponding elements of the two vectors together *if* their sum is greater than 3. Otherwise, the code should simply sum the elements.